

ATHINAI HELLINIKON, Greece

WMO: 167160

Lat: 37.8897N

Lon: 23.7417E

Elev: 142

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.4	38.5	20.9	15.8	42.5	24.7	18.9	44.0	24.3	48.3	21.9	49.4	5.8	350	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.1	96.5	70.1	93.8	70.1	91.7	70.1	76.3	88.7	74.9	87.6	73.6	86.0	10.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	119.1	84.2	70.8	114.0	83.2	69.4	108.7	82.2	39.9	88.9	38.5	87.6	37.3	86.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.8	16.7	31.4	101.3	3.6	2.4	28.8	103.0	26.7	104.4	24.6	105.7	22.0	107.4
			28.2	79.1	3.1	1.4	26.0	80.1	24.1	80.9	22.4	81.7	20.1	82.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.7	50.4	51.2	54.9	60.4	69.4	78.2	83.2	83.5	75.8	67.8	59.8	53.1
	DBStd	12.81	5.68	5.61	5.13	4.55	4.61	4.95	3.33	3.53	4.37	5.04	5.34	5.83
	HDD50	172	63	50	16	1	0	0	0	0	0	0	5	37
	HDD65	1882	451	387	314	149	13	0	0	0	0	32	167	369
	CDD50	5915	77	84	169	312	600	846	1031	1038	775	550	299	134
	CDD65	2149	0	0	1	10	148	396	566	573	325	117	12	1
	CDH74	21416	0	0	3	24	810	3936	6919	6941	2401	375	7	0
	CDH80	9027	0	0	0	1	149	1571	3280	3309	685	32	0	0
Wind	WSAvg	6.5	6.4	6.7	6.8	6.0	5.8	6.6	7.4	7.5	6.4	6.1	5.9	6.4
Precipitation	PrecAvg	14.2	1.9	1.6	1.6	1.0	0.6	0.3	0.3	0.2	0.6	1.7	2.3	2.6
	PrecMax	20.2	7.0	5.1	6.1	4.4	2.8	1.5	2.4	1.8	3.0	7.6	8.0	6.4
	PrecMin	6.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.0	1.5	1.2	1.3	1.0	0.7	0.4	0.6	0.3	0.8	1.7	1.8	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	65.6	71.3	78.2	87.7	97.1	99.8	100.5	92.1	83.2	75.1	68.4
		MCWB	57.5	56.6	58.7	62.0	65.2	69.1	70.8	69.7	69.7	69.3	64.9	61.9
	2%	DB	62.5	63.1	67.7	73.5	84.0	93.3	96.2	96.8	88.3	80.4	72.2	65.6
		MCWB	56.3	55.3	57.1	59.4	64.5	69.5	70.4	70.1	69.4	67.0	63.9	59.2
	5%	DB	60.8	61.4	65.3	71.3	80.9	90.7	93.6	94.2	86.2	77.9	70.2	63.8
		MCWB	55.0	54.4	56.0	58.8	64.4	69.1	70.2	70.3	69.1	66.3	62.9	57.8
	10%	DB	59.1	59.8	63.4	68.8	78.6	88.0	91.6	91.7	84.1	75.6	68.1	61.8
		MCWB	53.7	53.4	54.9	57.8	63.9	68.7	70.2	70.4	68.4	65.5	61.5	56.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	58.5	60.2	63.7	70.0	75.1	77.6	78.4	76.0	72.5	67.4	62.9
		MCDB	63.7	63.7	68.2	73.8	82.2	88.8	90.0	90.7	86.6	79.9	72.5	68.0
	2%	WB	57.3	56.7	58.4	61.9	68.2	73.3	76.0	76.8	73.7	70.4	65.4	60.4
		MCDB	61.3	61.5	65.6	70.9	79.5	87.0	88.9	89.0	83.2	77.2	70.5	64.7
	5%	WB	55.9	55.4	57.2	60.4	66.6	72.0	74.7	75.3	72.0	68.7	63.9	58.7
		MCDB	60.0	60.4	63.6	68.7	77.4	85.4	87.8	88.0	81.5	75.4	69.3	62.9
	10%	WB	54.2	54.1	55.8	59.0	65.3	70.7	73.5	73.7	70.6	67.2	62.3	56.8
		MCDB	58.6	59.0	62.0	67.0	75.6	83.8	86.4	86.2	80.5	73.8	67.4	61.0
Mean Daily Temperature Range	5% DB	MDBR	11.0	12.3	13.0	14.2	15.8	16.4	16.3	16.1	14.6	12.9	11.5	10.1
		MCDBR	11.9	14.1	15.1	16.4	18.4	18.7	18.7	18.4	16.5	14.7	12.8	10.7
		MCWBR	8.2	9.3	9.2	9.0	8.8	8.3	8.6	8.3	9.2	8.2	8.0	7.6
	5% WB	MCDBR	10.7	12.7	13.8	15.3	17.2	17.3	17.2	16.8	15.1	13.1	11.9	9.7
		MCWBR	8.2	9.2	9.2	9.3	9.4	9.2	9.7	10.1	9.5	8.6	8.3	7.7
Clear-Sky Solar Irradiance	taub	0.350	0.374	0.409	0.453	0.436	0.416	0.420	0.425	0.424	0.396	0.376	0.360	
	taud	2.433	2.349	2.247	2.130	2.242	2.335	2.333	2.309	2.312	2.369	2.392	2.407	
	Ebn at Noon	261	269	269	263	270	274	272	268	261	257	248	247	
	Edh at Noon	27	33	41	48	44	40	40	40	37	32	28	26	
All-Sky Solar Radiation	RadAvg	638	884	1327	1743	2084	2377	2416	2154	1649	1121	748	554	
	RadStd	70	104	105	148	102	118	52	45	87	89	73	73	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-148	N/A	N/A

Nomenclature: See separate page